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ORIGINAL PAPER

C-K design theory: an advanced formulation

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Abstract C-K theory is a unified Design theory and was first introduced in 2003 (Hatchuel and Weil 2003). The name "C-K theory" reflects the assumption that Design can be modelled as the interplay between *two interdependent spaces* with different structures and logics: the space of concepts (C) and the space of knowledge (K). Both pragmatic views of Design and existing Design theories define Design as a dynamic mapping process between required functions and selected structures. However, dynamic mapping is not sufficient to describe the *generation of new objects and new knowledge* which are distinctive features of Design. We show that C-K theory captures such generation and offers a rigorous definition of Design. This is illustrated with an example: the design of Magnesium-CO_2 engines for Mars explorations. Using C-K theory we also discuss Braha and Reich's topological structures for design modelling (Braha and Reich 2003). We interpret this approach as special assumptions about the stability of objects in space K. Combining C-K theory and Braha and Reich's models opens new areas for research about *knowledge structures* in Design theories. These findings confirm the analytical and interpretative power of C-K theory.

Keywords Design theory · Innovation · Creativity

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1 Introduction. C-K theory: initial reactions and issues raised

In this paper we present an advanced formulation of C-K theory, drawing on initial reactions to the theory and on new research findings. The new material helps clarify the unique properties of the theory and provides fruitful interpretations of the assumptions of other formal Design theories such as the Braha and Reich model (Braha and Reich 2003). Before outlining the issues discussed here, we begin with a brief overview of the premises of C-K theory.

1.1 A brief overview of C-K theory: modelling innovative design

C-K theory was introduced by Hatchuel and Weil (2003). It aims to provide a rigorous, unified formal framework for Design. It also attempts to improve our understanding of *innovative design* i.e. design which includes *innovation* and/or research as in the case of Science Based Products (Hatchuel et al. 2005). The name "C-K theory" reflects the assumption that Design can be modelled as the interplay between *two interdependent spaces* with different structures and logics: the space of concepts (C) and the space of knowledge (K). The structures of these two spaces determine the core propositions of C-K theory (Hatchuel and Weil 2003):

The structures of C and K Space K contains all established (true) propositions (the available knowledge). Space C contains "concepts" which are *undecidable*[1]

[1] A proposition is qualified as "undecidable" relative to the content of a space K if it is not possible to prove that this proposition is true or false in K. The notion of undecidability is well defined in number theory and in computing science (Turing's undecidability theorem).

propositions in K (neither true nor false in K) about *partially unknown* objects *x*. Concepts all take the form: "*There exists some object x, for which a group of properties* P_1, P_2, …, P_k *are true in K*". Design projects aim to transform undecidable propositions into true propositions in K. Concepts define *unusual sets* of objects called C-sets, i.e. sets of partially unknown objects *whose existence is not guaranteed in* K. During the design process C and K are expanded jointly through the action of *design operators*.

The design process and the four C-K operators Design proceeds by a step by step partitioning of C-sets until a partitioned "C-set" becomes a "K-set" i.e. a set of objects, well defined by a true proposition in K. This process requires four types of operators: C-C, C-K, K-K and K-C. These operators are explained later in the article. The combination of these four operators is a unique feature of Design. They capture all known design properties including creative processes and explain seemingly "chaotic" evolutions of real practical design work.

1.2 Issues raised about C-K theory

The first publication of C-K theory attracted interest from both practitioners (Fredriksson 2003) and scholars. In recent years, C-K theory has been introduced in several industrial contexts [most of these applications have been described elsewhere (Le Masson et al. 2006)], but in this paper we focus on the reactions to the theory in academic papers. Kazakçi and Tsoukas (2005) underlined the power of the theory when compared to other theories such as Gero's evolutionary design (Gero 1996) and suggested introducing the designer's *environment, E*. This extension does not change the basic assumptions of C-K theory but suggests a practical organization of space K that helps develop new types of personal Design assistants. Salustri (2005) sees C-K theory as a "*unique and interesting Design theory*" but asked for increased rigour in its presentation. He uses C-K propositions as an inspiring source for a new language of action logic for Design. In this language, the "concepts" of C-K theory are interpreted as the designer's dynamic "beliefs" concerning design solutions. However, Salustri found no necessity to assume C-sets in his model. Le Masson and Magnusson (2002) used C-K theory to enhance users' involvement in design. They interpreted the most surprising user ideas as *concepts* which deserve further design expansion with the help of experts. Ben Mahmoud-Jouini et al. (2006) also used C-K theory in addition to classic creativity techniques to build an innovation strategy in a car supplier company. Elmquist and Segrestin (2007) modelled creative drug design with C-K theory to enrich scouting and scanning methods for the acquisition of new molecules.

As well as confirming the potential of the theory, these authors and other readers (conference and journal reviewers, workshop participants etc.) pointed out a number of issues that were not sufficiently addressed in the previous presentation of C-K theory (Hatchuel and Weil 2003): what is the definition of Design in C-K theory? How is it related to the usual pragmatic views of Design? What are the main aspects of Design that C-K theory captures better than other theories, in particular recent Design theories such as those put forward by Braha and Reich (2003)? In this paper we discuss these issues and present new clarifications and findings that we hope improve on the first presentation of C-K theory.

1.3 Outline of the paper

The paper is divided into three parts. In Sect. 2, we evoke the "pragmatic" definition of Design as good mapping between required functions and selected structures. Design theories generalize this definition by describing *dynamic mapping*. However, dynamic mapping is not sufficient to describe the *generation of new objects and new knowledge* which are distinctive features of Design. We show that C-K theory captures such generation and offers a rigorous definition of Design. In Sect. 3, we show how the combination of four C-K operators enables reasoning on *unknown or changing* objects. This is illustrated with the example of the design of Mg-CO_2 engines for Mars explorations. In this case, Design not only maps functions and structures, it also shifts the identity of the engine and the type of missions it will serve. In Sect. 4, we use C-K theory to interpret Braha and Reich's topological structures (i.e. *closure spaces*) for design (Braha and Reich 2003). We show that these models assume the stability of objects in K. Combining C-K theory and closure spaces clarifies the distinction between rule-based design and innovative design. These results confirm the explanatory and interpretative power of C-K theory. We conclude (Sect. 5) the paper by indicating some areas of research opened by these findings.

2 The definition of Design in C-K theory

2.1 Pragmatic definitions of Design

Usual definitions of Design are *pragmatic descriptions* of a professional challenge (Evbwuoman et al. 1996). Designers receive a "brief" or "specifications" of a product (or service) from a customer and in return, they are expected to offer several "proposals" or "designs" which meet these specifications. A more realistic approach to Design acknowledges a continuous interplay between designers and customers. Specifications may change in reaction to

proposals or to unexpected problems discovered during the process. In this case, Design follows cycles of mutual adjustment between specifications and solutions until a final "solution" is reached. A large amount of research into engineering design does not require a more precise definition than this. Theoretical problems only arise when design itself becomes the object of academic inquiry (Evbwuoman et al. 1996; Blessing 2003; Simon 1979). Then, simple questions unveil difficult issues: is it possible to distinguish design improvements from technological improvements? How can we establish a design methodology without a rigorous definition of Design? What are the links between Design and innovation?

2.1.1 *Formal models of design: the limits of dynamic mapping*

These issues are crucial for researchers who work on design methodologies and/or mathematical representations of Design. However, even the most abstract Design theories draw on the same pragmatic definition of Design: Design is a *mapping* process between functions and design parameters or structures (Suh 1990; Yoshikawa 1981); this may be achieved in a small number of fixed steps (classic systematic design) or may follow a more evolutionary process (Gero 1996). Within the same perspective, Braha and Reich (2003) generalized Yoshikawa's Design theory and presented an encompassing model, the *Coupled Design Process* (CDP in this paper) that accounts for various properties of design including, non-linearity, non-optimality, conflicting goals and exploratory processes. In their approach, Design is modelled as a *dynamic mapping process* between a function space F (set of functions) and a structural space D (set of design options or parameters). A special form of this co-evolution is modelled with *closure spaces* which are an interesting way of describing *refinement steps for functions and structures (*In part 3, we discuss the interpretation of closure spaces with C-K theory).

However, is the pragmatic definition of Design a rigorous approach to design processes? And consequently, is dynamic mapping *sufficient to model* Design? The answer is negative, as we can find situations which require no design activity, but where dynamic mapping is nonetheless necessary. Moreover, dynamic mapping does not capture the main operations involved in design situations where new objects have to be generated.

2.1.2 *Dynamic mapping in problem-solving: the example of a lost driver*

Let us take the example of a driver lost in an unknown country. He is looking for a "*convenient hotel, not too far away and not too expensive*". The driver has no guidebook to the country and has to ask the people he meets for information to help him adjust his own desires to the solutions available. Herbert Simon (1979) often used similar situations to describe problem-solving procedures based on the dynamic fit between solutions and satisfaction criteria. However, the driver will not *design* the hotel where he decides to stay. We could say that he designs a *decision function* to find it; and Decision theory can be seen as a minimal form of design. Yet, Design usually involves far more than selecting existing solutions. Therefore, dynamic mapping is not a distinctive aspect of Design, and we need to identify the features of design that it fails to capture.

2.2 Design as the generation of new objects

Let us introduce example A, inspired by a real case study. We will use it in the following sections of the paper to illustrate the propositions of C-K theory.

Example A: designing an Mg-CO_2 engine for Mars exploration Future Mars missions face a well known energy problem. Spaceships have to transport all the propellant for the Mars exploration and the return journey; in view of the great distances involved, this is no minor issue. Given that Mars' atmosphere is made of CO_2, this could be a good oxidant for burning metals such as magnesium. Could it be possible to "refuel" with CO_2 on Mars? Scientists suggested the option of designing Mg-CO_2 engines for Mars missions.[2]

Example A introduces a common, yet distinctive, feature of Design. The lost driver had neither to *design* hotels nor *to make them exist*. He had to *find* and *choose* them. Mathematically, the driver problem can be approached by programming heuristics, problem-solving theory and multicriteria decision-making (Simon 1969). These models fully capture the dynamic mapping between solutions and criteria, but not the "*generation*" of new things, i.e. in example A, *the definition* of a new engine whose principles are not necessarily known today, as well as the identification of conditions guaranteeing *the existence* of such an engine. Hence, a complete definition of Design has to account for *two joint processes* that are not clearly outlined by the pragmatic definition:

- dynamic mapping between specifications and design solutions.
- The generation of objects unknown at the beginning of the process and whose existence could be guaranteed

[2] This case was developed using C-K theory by our student Michael Salomon during his Major course for the engineering degree at *Ecole des Mines de Paris* in collaboration with CNRS-LCSR. His work contributed to the material published in Shafirovich et al. (2003).

by knowledge that may be discovered during the process.

The combination of these two issues leads precisely to the premises of C-K theory.

2.3 The premises of C-K theory: meaning and role of "Concepts"

2.3.1 The logic of Design "briefs"

The starting point of a design project is described in pragmatic terms as a "brief", an "idea" or "abstract specifications". These expressions attempt to describe an object that is not completely defined and whose conditions of existence are not completely known. Therefore, the only way to start the design process is to formulate an *incomplete, even ambiguous group of desired properties* for this object. To capture the reasons and rationale for such odd formulations we need to model both what is known and what is partly unknown. The two spaces of C-K theory fulfil this need.

Definition of space K We assume an expandable Knowledge space K, which contains true *propositions* characterizing partly known *objects* as well as partly known *relations* between these objects. In K, all propositions are true or false. K is *expandable* i.e. the content of K will change over time and definitions of some objects of K may also change. In practice, K is the established knowledge *available* to a designer (or a design team). Conflicting views and uncertainties are also true propositions of K. In example A, K contains several knowledge bases: Mars science, combustion science, future Mars missions, Mars exploration politics and main actors.

Definition of space C and "Concepts" We consider propositions of the following type P: "*There exists some entity x (or a group of entities) for which series of attributes* A_1, A_2, A_k *are all true in K*". We define P as a *concept* relative to K if P is *neither true nor false* in K. We assume that Space C is expandable and contains all the concepts relative to K. Space C is a key premise in C-K theory. Its unusual structure controls the main properties of C-K theory and captures the core features of Design. It unravels the nature of briefs and allows new objects to be generated during the design process.

2.3.2 Why Design begins with a concept?

Concepts clearly capture the nature of briefs: either the brief is "undecidable" in K or the design process has already been completed. Concepts also confirm that ambiguity, ill-defined issues and poor project wording are not problems or weaknesses in design, they are necessary! Moreover, undecidability and incomplete concepts can be seen as consistent *triggers* once design is perceived as an *expansion* process (see below). For the same reasons, concepts are not propositions that can be tested like *scientific hypotheses*. As the latter have to be assumed as true this would mean that the design work has already been done. For instance, in example A, we cannot begin to design a new Mg-CO_2 engine for Mars exploration and immediately test it, but we can check whether *a design proposal is acceptable as a concept*.

Coming back to our Mg-CO_2 engine, let us consider the proposition C_0: "*There is an Mg-CO_2 engine that is more suitable to Mars missions than classic engines*". We then have to prove that it is a concept. Obviously, it was not possible to prove that C_0 was true with existing K, but was C_0 false in K? In fact, it needed only one proposition in K to "*kill the concept*". To meet the requirement of a good propellant, the combustion of Mg and CO_2 had to create sufficient "specific impulse" (i.e. energy for movement), otherwise there would be no engine at all. This property could be tested *without fully designing an engine* and was therefore assessed scientifically. This test simply proved that there was no proposition within existing K that proved that C_0 was true or false. Thus, C_0 was a suitable concept for further design. According to Pahl and Beitz's systematic design (1984) the main function of an engine is to produce sufficient energy; we therefore simply checked this function. Yet, Pahl and Beitz recommend modelling *all* the main functions in a first design phase, a task which was clearly impossible in this case. Moreover, the satisfactory level of specific impulse from a propellant's combustion can be interpreted as a function, as a conceptual model or even as an embodiment solution. This illustrates the ambiguity of classic design phases when design is innovative. C-K theory frees the designer from such predefined steps and categories. What counts is the consistency of the operations between C and K and the expansion produced in the process.

2.3.3 Design simultaneously expands C and K

The pragmatic view of design describes a dynamic mapping process between specifications and solutions. However, it is clear that this approach fails to account for the expansions occurring in space C and in space K during the actual process. Let us start a design process with a concept C: "*there exists an x with a set of attributes* A_0". At step i, the designer has changed the initial set of attributes A_0 into A_i by adding or subtracting new attributes and has introduced some partial design parameters D_i. At this stage, a new proposition C_i has been formed: "*There exists x with a set of attributes* A_i, *which can be made with a set of*

design parameters D_i". There are three possibilities for the logical status of C_i in K:

1. C_i is false in K and the design process has to change some of the A_is or the D_is;
2. C_i is true in K and (D_i, A_i) is one candidate as a "solution" for X; we call it a "*conjunction for x*";
3. C_i is neither true nor false in K: hence it is *a new concept* and we have to continue the design process.

In the two-first cases *we have added new propositions to K*; in the third, we have added a *new concept* to C. Thus design not only generates "solutions" but also, by the same procedures, *new concepts* and *new propositions in* K. It is therefore more rigorous to describe the design process as a *dual expansion* of spaces C and K. This finding can also be based on empirical observations. Design often generates knowledge that is finally used for a different purpose than the initial brief; or stops at *an intermediary concept* which can even be sold as such. For example, the designer of a movie may stop after writing the story and sell it to a film maker who will adapt it to suit his or her own views. Hence, the premises of C-K theory are both more rigorous and more realistic than the pragmatic definition of design.

2.4 Conclusion of Sect. 2: a definition of Design

All the premises and initial propositions of C-K theory are essential in formulating a highly precise, general definition of Design.

Definition Design is a reasoning activity which starts with a concept (an undecidable proposition regarding existing knowledge) about a partially unknown object x and attempts to expand it into other concepts and/or new knowledge. Among the knowledge generated by this expansion, certain new propositions can be selected as new definitions (designs) of x and/or of new objects.

This definition does not contradict pragmatic definitions of Design. It is more general and more complete. It introduces the generation of new objects and consistently defines the departure point for a design project. In the next section, we illustrate this definition *in action*, as all operations modelled by C-K theory can be deduced from these premises.

3 C-sets and C-K operators: expanding knowledge and revising object identities

Pragmatic accounts of Design portray the changing, often surprising paths followed by designers groping about a solution. C-K theory captures this process and explains its specific rationality and logic by analysing the simultaneous expansion of C and K. However, space C and space K follow two different, albeit interdependent, expansion patterns. We begin by examining the specific role of space C as it supports the logic of the whole process.

3.1 A central property of C-K theory: revising the identities of objects in Space C

Identity of an object in K Let us assume, in space K, propositions about a collection of objects O which *all* possess an attribute A_0 (example: "all known car tyres are made of rubber"). Thus, A_0 ("made of rubber") can be considered as a partial element of *the identity* of O. Let us put forward the proposition Q: "*There exists O without* A_0" ("there exist car tyres without rubber")". If K contains a *universal proposition* which says that all O, whatever the time or place, have the attribute A_0, then Q is false. But if K only contains the proposition: "*All known Os have the attribute* A_0"[3] then Q is a potential concept that may lead to a *revision of the identity* of O. As C allows for such potential changes in the identities of objects in K, C-K theory therefore captures *the birth of new objects*.

This property of Space C was not emphasized sufficiently in the first presentation of C-K theory (Hatchuel and Weil 2003). It highlights the key importance of space C and clarifies the power of design reasoning. This property that we call "*power of expansion*" is, to the best of our knowledge, a unique way of capturing creativity or invention *within* Design theory and not as an *external addition*. However, this power of expansion depends on particular conditions in K Whenever possible, *universal propositions* should be avoided in K as they are logical obstacles to the revision of object identities. Thus C-K theory supports the intuitive notion that Design is not very consistent with universal, fixed object identities. The formulation of undecidable propositions concerning partially unknown objects obviously requires some precautions and we therefore introduce the notion of *concept-sets*, or C-sets, which are a powerful analytical tool.

3.2 Concept-sets as sets of partially unknown objects

In space C, we define *concept-set* as follows: a set defined by a proposition which is a concept relative to K. For example, if C is the concept "*there exists an x with A(x)*", the C-set is the set of all objects x that verify A. C-sets present surprising properties. They are *neither empty nor non-empty*. This result is a corollary of the definition of a

[3] For example usual major premises in syllogism as "all humans are mortal".

concept. To prove that C-sets are non-empty, the only way is to exhibit an *x* verifying A in K. But this would mean that C is true in K, hence C is not a concept. The same type of proof can be used for the "empty" case. What is the meaning and role of C-sets? In classic programming theory or problem-solving theory (Simon 1969; Simon 1979; Simon 1995), the task is to explore a *problem space* containing a list of potential or approximate solutions. All solutions may not be accessible; it is however assumed that solutions are built by the combination of well defined objects like, for example, in the game of chess. In contrast, Design faces situations where *it is not possible to define even an infinite list of known design candidates or even to define what such candidates are*. C-sets capture this situation by modelling collections of *partially unknown objects* which verify a proposition which is *undecidable in K*. In example A, the set of "all Mg-CO_2 engines for Mars explorations" is clearly a C-set. It is not only impossible to list all possible Mg-CO_2 engines, but the design parameters of such engines are also partially unknown when design begins. C-sets are special sets which, to our knowledge, have not been described in the Design literature to date. To rigorously define C-sets, we make some restrictions to the standard axioms of set theory.

Axioms for defining and partitioning C-sets C-sets are defined within a restricted axiomatic of Set theory. Namely ZF (Zermelo–Fraenkel) without two important axioms: the axiom of choice (AC) and the axiom of regularity (AR) also known as axiom of foundation (every non-empty set A contains an element B which is disjoint from A).[4] This axiomatic of Set Theory is described as ZF-non AC, -non AR. Axiom of choice and axiom of regularity are respectively the warrantors of the existence and selection of one element in a set (Jech 2002). As C-sets are neither empty nor non-empty, they cannot verify these axioms. These axioms are usually formulated on the condition that the set is non-empty, a condition that we can neither accept nor reject for C-sets (Jech 2002). Although some authors (Salustri 2005) do not see the need for the axiomatic of C-sets, we stand that it captures the neglected, yet crucial, fact that during the design process we manipulate collections of *objects* which do not have *operational and stable* definitions. Designers work with sketches, models or mock-ups which are actual *representations* of a family (often infinite) of future objects which are still partly unknown and related to undecidable propositions. They cannot logically extract and manipulate a single, well defined design solution *until it has been decided conventionally that design has ended.* These families of representations have the properties of C-sets.

The axiomatic of C-sets explains the structure of expansion of space C. As shown in Hatchuel and Weil (2003), due to the rejection of the axiom of choice and axiom of regularity, the *only* operations allowed on C-sets are *non-elementary partitions (or inclusions)*. These partitions are core operations of C-K theory. Design can only partition an initial concept in the hope that this expansion of attributes will create useful new concepts and new knowledge. The partitioning attributes in C must be extracted from K. In return, K is expanded by attempts to check the logical status of propositions. Four operators (C→C, C→K, K→K, K→C) produce these expansions which transform C into K and conversely. This C-K interplay is illustrated below with a summary of the Mg-CO_2 case. We underline how C-K operators organize the design process and also allow for a *flexible, changing definition of objects.*

3.3 The operators of C-K theory: an illustration with example A

Having assessed that "*there exists an Mg-CO_2 engine for Mars exploration…*" was a concept (see Fig. 1), the next stage is to partition this concept in space C.

3.3.1 Phase 1: partitioning with known Mars missions

What was known about Mg-CO_2 engines in K? That they should perform better than classic ones. And about Mars missions? The available options where found (C→K) in the previous Mars missions simulation and the validation tools of the Space Agency concerned. Partitioning with each mission scenario (K→C) generated Mg-CO_2 concepts that could be compared to other propellants *without further descriptions of the engine* (K→K). However, it was found

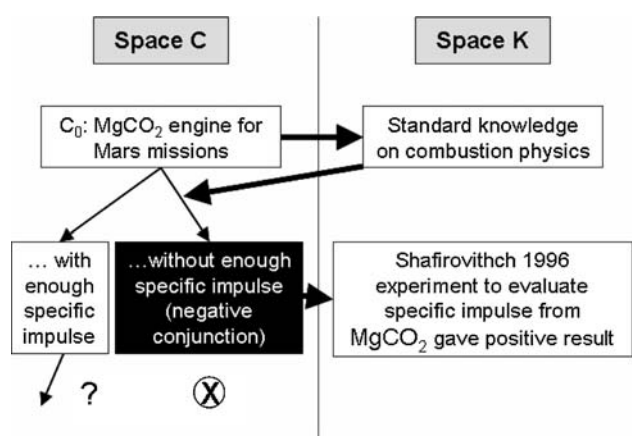

Fig. 1 Assesing a Concept of Mg-CO_2 engine

[4] The rejection of the axiom of foundation was not mentioned in Hatchuel and Weil (2003). It was suggested to us by our student Mathieu le Bellac in his minor dissertation for the Master in management (MODO) at Université Dauphine.

that if usual mission criteria were maintained, no Mg-CO_2 engine would globally perform better than standard propellants! In other words, for all known mission scenarios added to C_0, the new proposition was false in K. To carry on the design process new partitions of C_0 were needed (i.e. partitioning the box "other?" in Fig. 2). Meanwhile, what happened in K? The scenario analysis had created new and unexpected knowledge. It appeared (K→K) that each time Mg-CO_2 engines were used *only on Mars* the mission performed better than others with classic criteria. This new proposition in K (see Fig. 2, the black block with white letters in K) offered a new "expanding" partition (see below).

3.3.2 Phase 2: revising the identity of the engine

This new proposition suggested (K→C) a new concept: "*there exists an Mg-CO_2 engine used only on Mars during Mars explorations*" (see Fig. 3). Once again, how could we partition this new concept? Could we expand the knowledge available on the missions performed on Mars (C→K)? The question stimulated additional research (K→K) which showed that existing mission scenarios poorly modelled activities that could be performed on Mars. *The rover solution was too implicit in existing definitions of missions to perform on Mars*. Instead, a new typology of missions was established with new models of *mobility,* new *scientific experiments*, new *communication tasks, etc.* This new knowledge on Mars exploration generated new partitions for C. For example, rapid refuelling of CO_2 for unplanned moves (see Fig. 3) in case of environmental dangers (dramatic storms are common on Mars) was a new potential attribute of the engine. At that stage, with a new concept such as *"an Mg-CO_2 engine, only used on Mars for a new type of mobility that could be either planned or unplanned*", the identity of the designed object *was shifting*. The first concept was evaluated as a complete

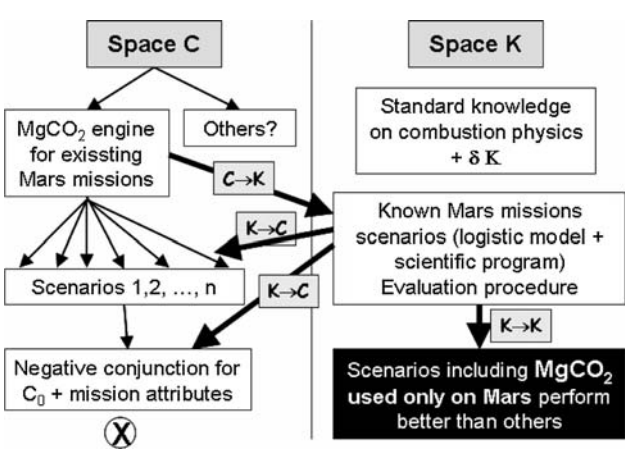

Fig. 3 Revisiting the identity of the engine

alternative to existing propellants. The new concept of "an Mg-CO_2 engine" was now associated with a wide variety of movements on Mars which evoked a *new type of vehicle* for Mars exploration: a "hopper" (see Fig. 4) (Shafirovitch et al. 2003). It is worth mentioning here that this *identity shift* is captured by a group of partitions that could not be activated at the beginning of the process.

3.3.3 Phase 3: designing for prototyping

Thus, a new concept for the engine led to the definition of a new concept of vehicle, and large amounts of new knowledge about Missions on Mars were then generated. What was the next step (C→K)? The standard knowledge was that "An Mg-CO_2 engine for a Mars hopper" should be testable by earth prototyping". But which prototype should be designed? Answering this question meant searching (K→K) for testable conditions (K→C) that would partition the concept of an Mg-CO_2 engine for a Mars hopper. These conditions were obtained by a computation tool (K→K) that defined mass limits for the engine and its associated CO_2 plant. This introduced a new

Fig. 2 Attributing known missions

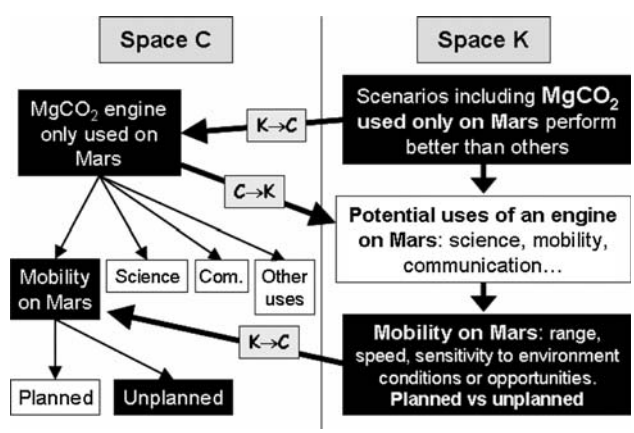

Fig. 4 Designing for prototyping

proposition in K: "*an Mg-CO_2 engine for a Mars hopper that enables extended mobility and unplanned movements has an engine mass and a CO_2 plant mass limited to a defined domain*". This clarified the conditions for the design of a new prototype: such demonstrator should help to check whether the design domain in question was a killer criterion for the engine concept. The following partitions were all oriented towards the design parameters of the prototype.

Example A has been described in more detail in Hatchuel et al. (2004). It has also been modelled by Salustri (2005). The above overview illustrates an important property of C-K theory: a small number of operators capture *the generation and changing identity of an object*, a complex process which would seem "chaotic" if C and K were not modelled simultaneously and interdependently.

3.4 A summary of C-K operators

We shall now summarize the specific functions of the four operators illustrated in example A.

3.4.1 The four C-K operators

- C→K operators search *attributes* in K which can be used to partition concepts in C.[5] They also contribute to the *generation of new propositions in K*. Each time a concept C_0 is modified by a new attribute we must check whether the new proposition is still a concept. This does not simply involve answering 'yes' or 'no'. New propositions are generated that may be new sources of attributes for the following partition (this is what happened for the Mg-CO_2 engine mission tests). Thus concepts have *an exploratory power* in K through their own validation.
- K→C operators have symmetrical functions to the previous ones. They generate tentative concepts by assigning new attributes. They also assess the logical status of new concepts and maintain the consistency of the expansion of C.
- C→C has been seen as a *virtual* operator (Kazakçi and Tsoukias 2005) as the main operations travel through K. In fact, it is of utmost importance in the formation of the results of a C-K process. "Design solutions" are *chains of attributes that contains C_0 and form new truths in K*. Hence, C→C operators are graph operators in Space C that enable the analysis of chains, paths, sub-graphs, and so on.
- K→K operators encompass all classic types of reasoning (classification, deduction, abduction, inference, etc.). Moreover, any design methodology that can be performed as a program (or an algorithm) without any use of concepts and C-sets is finally reduced to a K→K operator (for example, the genetic algorithm for optimizing an engineering system uses only standard calculus and logics).

The structure of these operators once again underlines *the major role of space C*. It gives birth to *three new operators* which do not belong to classic modes of reasoning. This is a new confirmation of the *specificity* of Design compared to other modes of reasoning which can be described using only K→K operators.

3.4.2 The asymmetric structures of spaces C and K

These operators generate two different yet interdependent structures in Space C and Space K. In C we can only partition C-sets as no other operations are allowed. Hence, C is always *tree-structured* and presents a divergent combinatorial expansion, whereas K is expanded by new propositions that have no reason to follow a stable order or to be connected directly. As suggested by Fig. 5, K grows like an *archipelago* by the adjunction of new objects (new islands) or by new properties linking these objects (changing the form of the islands). The complete mathematical treatment of these properties is not straightforward. It is beyond the scope of this paper and will be treated in forthcoming papers.

3.5 Synthesis: expanding partitions and the changing identity of objects

C-K operators simultaneously model dynamic mapping and the distinctive feature of Design: *the generation of new objects*. This is achieved by the specific logic of C and the interplay between C and K. If we are limited to K-K

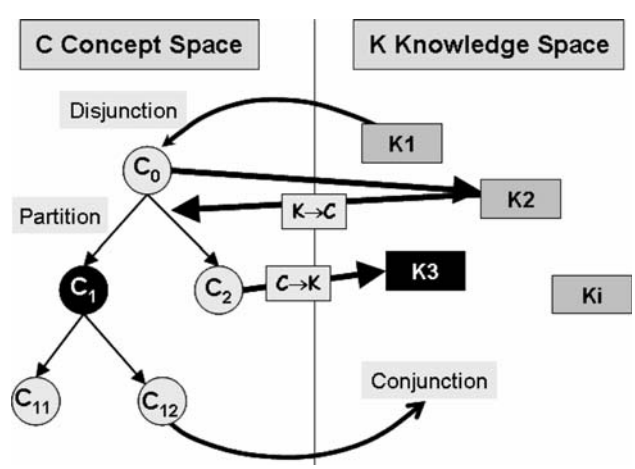

Fig. 5 Asymmetric structure of spaces C and K

[5] It should be noted that subtracting an attribute is equivalent to adding the negation of this attribute.

operators, we can prove theorems and simulate dynamic mappings, but the definition and identity of known objects remain stable as long as no paradox or contradiction appears. Thanks to Space C, we capture a more flexible logic. Given any object O, we can generate a concept Co if we are able to formulate an *undecidable* proposition in K. The key mechanism of this undecidability is the addition of an attribute to C_0 which is *not part of the existing knowledge* about O in K. For instance, "There exists a wireless home TV" would be a potential concept if "wireless" was neither a known attribute of existing home TVs, nor an attribute forbidden by existing knowledge. This would be an *expanding partition of Home TVs*. However, the same attribute ("wireless") is a "*restricting partition*" for phones, as mobile phones are well known to us. Expanding partitions are possible only in C, where they help to formulate concepts. They are the instruments which generate new objects, and C-K interplay is the source that provides new potential expanding partitions. More profoundly, expanding partitions reveal the *incompleteness* of K about O or the degree of "unknownness" of O in K. They are also powerful analytical tools for the study of other Design theories.

4 The interpretative power of C-K theory: a discussion of Braha and Reich's topological structures for Design

In this section we underline the interpretative power of C-K theory by analyzing a Design model proposed by Braha and Reich (2003), the *Coupled Design Process* (CDP).[6] According to the authors, CDP is more general than Yoshikawa's General Design Theory (Yoshikawa 1981; Reich 1995). We do not discuss this issue here, but simply establish that interpreting CDP with C-K theory highlights the meaning of the topological assumptions of CDP and opens new paths for further research.

4.1 Overview of CDP: modelling with Closure Spaces

CDP maintains the pragmatic distinction between a space of functions F and a space of structures (or design solutions) D; $F \times D$ is called the *Design Space* and an element $<f, d>$ of the design space is called a *design description*. The designer is assumed "*to start with an initial description* $<f_0, d_0>$". He then transforms this description through a sequence of $<f_i, d_i>$s; each transition is interpreted as "a *simultaneous refinement*" of the structural and functional solutions. Moreover, to cast these transitions more formally, the authors suggest a specific topological structure for F and D based on *closure spaces*. It is assumed that in F (or in D) there is a list of functions which presents a specific *order structure:* between two functions f_i, f_j there is an *order relation*: f_i "*is generated by*" f_j, which means that f_j *refines* f_i. The closure of a function f_0 is the list of functions that "generates" f_0 (or "refines" f_0).

All these structures allow the authors to define a finite sequence of refinements of either functions or structures which generate a possible dynamic mapping process for the designer: "*the designer starts with a candidate design solution* do *that needs to be analyzed, since its structural description is not provided in a form suitable for analysis. To overcome this problem the designer creates a series of successive design descriptions such that each design description in this "implication chain" is implied by the design description that precedes it*" (Braha and Reich 2003) (p.191). Design stops when the mapping is successful or when no refinement is possible and "*this situation can trigger the knowledge process in an attempt to continue the refinement process*."

CDP and C-K theory have many similarities. They both describe a dynamic refinement process. However, interpreting CDP with C-K theory highlights the implicit assumptions of CDP on three important issues: the departure point of a design process, the meaning of closure spaces and the "refinement" model.

4.2 The initial proposition of a design process

The departure point of CDP is defined with vague formulations. The authors describe $<f_0, d_0>$ as an "*abstract formulation*, a "*first idea of a solution from the designer*" that is still incomplete and ill-defined. Yet, they do not discuss the status of $<f_0, d_0>$ in relation to existing closure spaces of F and Ds. Two additional assumptions are necessary to clarify the status of $<f_0, d_0>$:

1. $<f_0, d_0>$ is not contradictory to what is known about the closure of $F \times D$;
2. $<f_0, d_0>$ is not a direct deduction of a subset of the closure of $F \times D$, otherwise the design work has already been done.

Without such assumptions CDP cannot easily assess whether $<f_0, d_0>$ is really a design problem. From the point of view of C-K theory, the first step would be to check whether $<f_0, d_0>$ is a *concept* within *existing* knowledge and to prove the undecidability of $<f_0, d_0>$ in K. This leads to the reverse question: what are the topological structures of $F \times D$ that make a proposition such as $<f_0, d_0>$ undecidable i.e. *neither implied by* these structures, nor forbidden (made false) by them? This remark is typical of how C-K theory can stimulate new research in the direction opened up by Braha and Reich.

[6] The acronym CDP is not mentioned by the authors, but is used here for the sake of concision.

4.3 The topological models of functions and structures: rule-based design and stable object identities

CDP models a refinement process of functions or structures with topological structures describing order relations. These assumptions can be interpreted as specific, stable properties of certain objects. In the language of Computer Science or Artificial Intelligence, closure spaces capture knowledge structures, generally referred to as "object models" (Abadi and Cardelli 1996). Our interpretation is confirmed by the car design example used by the authors. They describe the car as an object for which the available knowledge is modelled by standard production rules (if A then B). Design reasoning is thus equivalent to an expert system using forward and backward rule activation. More generally, assuming stable closure spaces can be interpreted as assuming stable object identities. To say that f_i *is generated by* f_j (or f_j *refines* f_i) is equivalent to saying that there exists an object "O" such that *if* f_j *is true for O then* f_i *is true for O*. The authors clearly acknowledge this interpretation as they establish a clear equivalence between rule-based design and stable closure spaces. Therefore, according to the topological assumptions of CDP, Design is a program which aims to combine existing objects that can be described in varying detail. The task of the designer is therefore to look for successful mappings, using increasing levels of refinement. However, no new objects can be generated if the refinement is always controlled by *pre-established closure spaces*.

This limitation disappears with C-K theory. Functional and structural closure spaces are considered as transient propositions in K, while partitions in C attempt to reshape *closure spaces* in K. Braha and Reich's topological structures can even be used as *an interesting design test:* the degree of revision of *F* or *D* closure spaces can be seen as an indication of the degree and extension of innovativeness of a design. In the case of the Mg-CO_2 engine, the function "mobility on Mars" was initially modelled by a closure function space that *was restricted to standard known* missions implicitly linked to the "rover" solution, a closure in the design parameters space. This confirms the need to study not only the *F* and *D* closures but also the $F \times D$ topological structure, at least to avoid *an implicit dependency between functions and structures that could be hidden by the separate closures*. C-K theory avoids this classic design trap by allowing for the revision of existing closure spaces.

4.4 Closure spaces and expanding partitions

Braha and Reich mention the important trap of "*poor quality knowledge*" that can lead to *"potentially exploring only inferior parts of the closure, leaving out the more promising solutions*". Yet, without explicit modelling of a space of knowledge, this type of judgement on the available knowledge is not modelled in the theory. Instead, if we assume that closure spaces are always K-dependent, *innovative design* can be approached by the following issue: how can we revise an initial closure space during the design process? Within C-K theory the answer is straightforward: the regeneration of closure spaces can *be directly linked to expanding partitions*. These partitions do not *refine* a function or a structure, otherwise they would be *restricting* partitions. Instead, the former partitions expand a concept and/or generate new knowledge that can change the boundaries and content of closure spaces. Describing the refinement process of a functional space, Braha and Reich remark that it can lead to a special list of functions that *does not belong* to the closure space : "*specification lists that are not included in F and such that each one generates specification lists in F*". In our view, this remark precisely describes a meta-structure connecting closure spaces in K. The authors associate such meta-structures with collaborative design[7] where designers share their colleagues' knowledge. However, more generally speaking, we can view any knowledge space K as a composition of *partly connected multiple transient closure spaces*. The task of expanding partitions is precisely to generate new connections which will prepare for the progressive *reshaping* of the closure spaces. This is exactly what is captured by C-K theory. In return, the closure space model confirms that expanding partitions are not "refinements". It also helps to understand that the dual expansion of C and K changes the definition of objects by allowing the reshaping of implicit closure spaces that may act as initial patterns in K.

Finally, this new perspective on the topological structures proposed by Braha and Reich (2003) does not refute the value of these structures in terms of modelling. On one hand, the notions of C-K theory (mainly concepts and expanding partitions) clarify the assumptions behind these topological structures. On the other hand, such topological structures can be seen as *interesting yet specific models* of the content of space K. Closure spaces can capture GDT, rule-based design and machine learning heuristics. Thus, by combining the two theories, we can establish highly general and powerful propositions:

Proposition 1 When space K is only defined by stable, separate, closure spaces, then C-K theory and CDP describe similar processes, and Design can be modelled by Knowledge-based and learning algorithms.

[7] We can also recognize a meta-structure in the logic for "infused design" proposed by Shai and Reich (2004a, b), a model for the aggregation of several knowledge bases in order to support collaborative design.

Proposition 2 If space K is described by transient closure spaces and by meta-structures linking these closure spaces, then C-K theory predicts that innovative design solutions (conjunctions in K) are always linked to a regeneration in the closure spaces.

5 Conclusion

In this paper we have made several steps towards an advanced formulation and the validation of the specific properties of C-K theory. The main results are as follows:

Design is not only a dynamic mapping process between functions and solutions. Design theory also has to describe the generation of new objects. Crucial elements of C-K theory capture this logic. The undecidability of concepts operationalizes the specific nature of design situations and explains the rationality of "briefs". Therefore, Design cannot be simply described as a problem-solving procedure. It is captured far better by the dual expansion of two different cognitive regimes: the *flexible approach* of C and the *truth-oriented logic* of K. As C-K theory accounts for this specific logic of Design, it provides a formal definition of Design which makes up for the shortcomings of pragmatic definitions of Design.

If Design is both a dynamic mapping process and a generation process for new objects, it requires four C-K operators as models of thought. Design theory extends known models of thought by introducing new analytical tools such as concept-sets based on "K-undecidable" propositions. Without such tools, Design theory is simply reduced to standard models of thought (K-K operators). By introducing these reasoning instruments, we have by no means fully modelled imagination, creativity or even serendipity. But at least C-K theory offers a framework that rigorously includes a key feature of innovative design: namely, the revision of the identity of objects and the possibility of expanding partitions.

The high generality and the modelling capacity of C-K theory are powerful instruments for the interpretation of other Design theories. Our discussion of Braha and Reich's topological structures is an example of this interpretative power. C-K theory helps to identify closure spaces of F and D as assumptions about the stability of objects in space K. This stability is consistent with rule-based design. Simultaneously, the strong propositions made by Braha and Reich can be used in combination with C-K theory to offer new propositions at a level of generality that is seldom reached in Design. This confrontation should be fruitful for both theories.

A variety of research issues can now be examined as a result of this progress in the consolidation of C-K theory.

C-K theory and topological structures of knowledge: the discussion of Braha and Reich's work calls for a systematic characterization of different types of structures in Space K and the corresponding Design theories that these structures allow. For instance, if closure spaces support rule-based design, which structures of K are consistent with systematic design *or different degrees of innovation* in the revision of objects? As we mentioned earlier, we must avoid universal propositions that rigidify the identities of objects. In this perspective, Doumas (2004) suggested exploring the type of design that would be predicted by C-K theory with a model of Knowledge built on "fluid ontologies" as proposed by Hofstader (1995). Such ontologies could be interpreted as fuzzy definitions of objects or even fuzzy closure spaces; however, additional research is required to establish this sort of equivalence.

C-K theory and research on creativity: In the past decades, engineering design literature has mainly borrowed results from the literature on creativity. There is now a fresh, stimulating opportunity: to explore how C-K theory could contribute to the field of Creativity. Ben Mahmoud-Jouini et al. (2006) and Elmquist and Segrestin (2007) used C-K theory to model creative processes in industrial R&D contexts. Such encouraging empirical results will be consolidated at a more theoretical level.

These research issues will be addressed in the future. In forthcoming papers we shall also back up these findings with a more complete presentation of the mathematical foundations of C-K theory.

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